

DRAW THE CHART

*A Chart-Native Price-Path Prediction Game
with Field-Relative Scoring*

Whitepaper — Version 3.0
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This document describes the product design, scoring architecture, and economic model of Draw The Chart (DTC). It is a concept and design paper: parameters given here are calibrated defaults and may be revised before and during live operation, with all changes versioned and announced. Nothing in this document is investment advice or a solicitation to participate in any regulated product in any jurisdiction where such participation would be restricted.

1. Abstract

Draw The Chart is a house-versus-player crypto prediction game in which the player expresses a market thesis by **drawing the expected future price path** on the chart itself — rather than choosing a binary outcome or opening a leveraged position. The player sees BTC history up to a locked anchor, draws an expected path over a fixed horizon (15 minutes to 7 days), stakes an amount, and is paid according to the quality of the prediction.

The defining feature of DTC is its **field-relative scoring architecture**. A drawn prediction is not graded on an absolute scale; it is ranked against a *synthetic field* of roughly 1,500 zero-information forecasts — random walks, flat lines, trend extrapolations, mean reverters, and bootstrapped historical paths — generated deterministically from the round’s cryptographic commitment and scored against the same realized market path. The player’s result is a percentile: “*you beat 87% of the field.*” The payout multiplier is a published function of that percentile, constructed so that its average over the percentile range equals exactly $1 - h$, where h is the house edge.

This architecture yields three properties simultaneously that absolute scoring systems cannot offer: **fairness** (round difficulty is normalized automatically, because the field faces the same market conditions as the player), **legibility** (a percentile against a visible field of competitors is the most intuitive score that exists), and **solvency** (a player with no predictive skill earns a uniformly distributed percentile, so the house edge holds exactly, by construction, in every market regime).

2. Motivation

2.1 The behavior already exists

Crypto users already perform the core DTC action informally. They draw trend lines and target paths, post “BTC to here first, then retrace” scenarios, and argue about timing and structure — not just direction. The behavior is already a game; no product captures it natively.

2.2 The gap in existing products

Existing products either compress the thesis or complicate it. Prediction markets reduce a path thesis to yes/no: “rallies first, fades, finishes slightly above” becomes “above X by date Y?”, and the nuance is discarded. Perpetual trading captures more, but demands leverage selection, position sizing, and liquidation management — a thesis can be right and still lose to execution mechanics. Casino-style products offer convex payouts and immediacy but remove analytical agency.

Category	Strength	Weakness
Prediction markets	Clean event resolution	No path expression
Perpetual / margin trading	Full expressiveness	Execution risk dominates the thesis
Casino-style games (e.g. Crash)	Excitement, convex upside	No analytical skill expression
Social charting	Rich expression	No economic closure

DTC occupies the gap between these categories: house-versus-player simplicity, chart-native input, deterministic and reproducible scoring, bounded downside, convex upside, and replayable outcomes. The chart is not a theme — it is the mechanic.

3. Product Overview

3.1 Definition

DTC is a house-versus-player BTC price-path prediction game in which users draw an expected future path and receive a payout based on how much of a simulated field of naive forecasts they outperformed.

3.2 The core loop

- Select BTC and a timeframe (15m, 1h, 6h, 24h, 7d).
- View history up to a locked anchor; the round is committed cryptographically.
- Enter a stake and draw the expected path; submit.
- The horizon elapses (or replays instantly in sandbox mode).
- Reveal: the realized path animates in, the synthetic field renders as a translucent swarm, and the player's rank within the field counts up.
- Payout: percentile → multiplier → settlement. The player can inspect the field paths that beat them.

The reveal is the signature product moment. “You beat 1,289 of 1,500 simulated traders — top 14% — 1.66×” is more legible, more credible, and more shareable than an opaque score.

3.3 Design principles

- **Expressiveness over compression.** Direction, timing, magnitude, and structure are all expressible in one stroke.
- **Determinism over mystique.** Every number on screen is reproducible from public code, the committed round payload, and realized prices.
- **Fairness is perceptual, not just statistical.** If a drawing that looks better scores worse, the system is wrong even if its math is internally consistent. This is an acceptance-tested requirement.
- **The house edge is a constant, not a hope.** Economics hold by construction under a zero-skill null; skill exposure is measured and bounded separately.
- **Bounded downside, convex upside.** Known floor, published curve, capped maximum.
- **Chart feel is product-critical.** The drawing interaction is the product.
- **Narrow first.** BTC only, five horizons, until fairness and economics are proven with real users.

4. Game Mechanics

4.1 Drawing input

The player draws a continuous freehand line over a fixed future zone to the right of the anchor. The raw stroke is normalized into a constrained prediction shape before scoring: a fixed number of control points per horizon, strictly increasing time coordinates spanning the full horizon, a minimum time spacing between points, and a maximum slope bounded by historical BTC movement. This keeps the interaction fluid while the scored object remains well-formed and resistant to degenerate inputs (Appendix C).

4.2 Path representation

All scoring is performed in log-return space, anchored at zero at the round start:

$$r(t) = \ln(P(t) / P_o), \quad r(T_o) = 0$$

Log returns make paths comparable across price levels and horizons. Both the predicted and realized paths are resampled to $N = 120$ evenly spaced points by linear interpolation, so every horizon is scored under an identical representation. Technical-analysis overlays (moving averages, bands, levels) are

available for player interpretation but are never part of the scored path.

5. Scoring Architecture

Path-similarity metrics measured on an absolute scale are inherently regime-dependent: the same forecast quality produces different scores in calm and in volatile markets, because every error term is implicitly compared against conditions that change each round. Extensive backtesting across five years of BTC data (2.7 million one-minute candles, 2021–2026) showed that no fixed parameterization of an absolute similarity score can be simultaneously fair across regimes and economically stable. DTC therefore separates the problem into three layers: **similarity** (how close was the drawing, perceptually), **ranking** (how good is that closeness relative to what was achievable this round), and **payout** (what does that rank pay). Difficulty normalization and house economics are handled structurally rather than by parameter tuning.

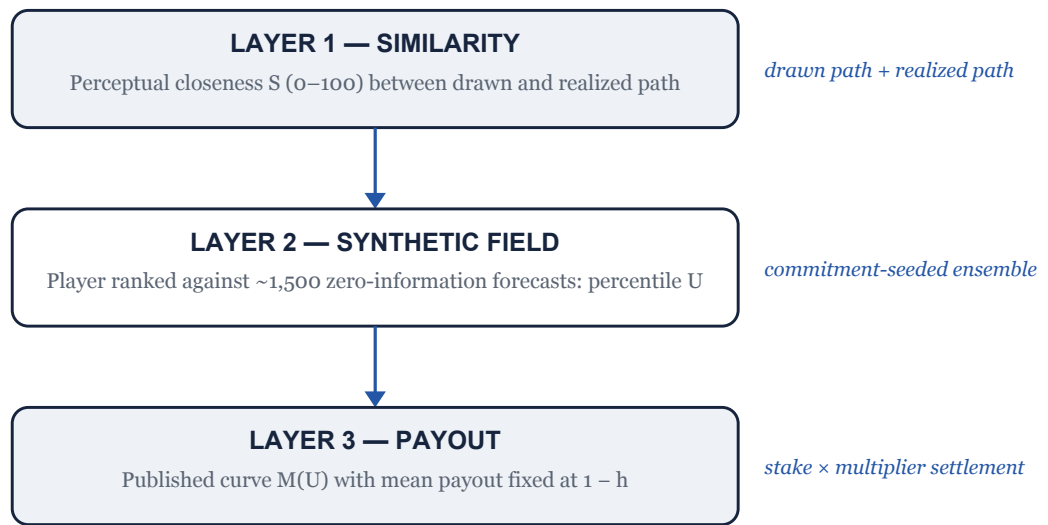


Figure 1 — The three-layer scoring architecture.

5.1 Layer 1 — Similarity

Layer 1 produces a perceptual closeness measure S (0–100) between the drawn path and the realized path. Its design goal is agreement with the human eye: tolerant of small timing errors, smooth everywhere (no scoring cliffs), and free of double-counted penalties. It is composed of three components.

Shape & timing (50 points). Banded dynamic time warping (DTW) — the formal version of what the eye does when comparing two charts. The two paths are aligned allowing local timing shifts of up to 10% of the horizon (Sakoe–Chiba band, $w = 12$ of 120 samples); the remaining distance D is normalized and mapped through a smooth exponential:

$$S_1 = 50 \cdot \exp(-\lambda_s \cdot D / (N \cdot \sigma_{eff}))$$

A forecast with the right structure that is slightly mistimed loses credit continuously, in proportion to the mistiming — never discretely. There are no “turn objects” to hallucinate or miss.

Direction (30 points). Multi-scale net-move agreement. The horizon is split into halves, quarters, and eighths; in each segment, the sign of the predicted net move is compared with the realized net move, and each segment is weighted by the size of the realized move, so segments where the market barely moved contribute almost nothing. Coarser scales carry geometrically more weight (decay 2.5).

Level (20 points). Bias (mean signed error) and endpoint error, each normalized by an effective volatility $\sigma_{\text{eff}} = \max(\sigma \text{ realized}, 0.5 \cdot \sigma \text{ typical for the timeframe})$, mapped through a smooth exponential. The hybrid floor keeps displayed accuracy stable in unusually quiet rounds.

Three rules are binding on Layer 1: no hard thresholds, no double-counting (each error type is penalized in exactly one component), and component correlation below 0.7 under the audit harness. A turning-point matcher (Hungarian algorithm over detected extrema) is retained purely as a reveal-screen explainer — “you called this top 9 minutes early” — it annotates, it does not score. Layer 1 has no direct economic role: money flows through the percentile, so its parameters are tuned for human agreement without ever touching the house edge.

5.2 Layer 2 — The synthetic field

For each round, $B = 1,500$ baseline paths are generated from a documented field generator conditioned on the anchor market state, scored with Layer 1 against the same realized path, and ranked together with the player. The player’s percentile is the mid-rank:

$$U = (\#\{\text{field below player}\} + \frac{1}{2} \cdot \#\{\text{ties}\}) / B$$

The field is a portfolio of **every strategy a player without market insight could play**. This is the anti-exploit core: if a naive strategy is in the field, playing it earns a near-uniform percentile and therefore exactly the house edge.

Share	Generator	Description
40%	Conditional block bootstrap	Blocks of recent 1-minute returns (trailing 30 days), rescaled to EWMA volatility at the anchor; carries realistic fat tails and clustering
20%	Geometric Brownian motion	Zero-drift GBM at EWMA anchor volatility
15%	Trend extrapolators	Lookback drift continued over $0.5 \times 1 \times 2 \times$ windows, damped and overshoot variants, with noise
10%	Mean reverters	Pull toward the lookback mean at several speeds, with noise
10%	Flat & drift lines	Flat line and small end-drift variants
5%	Smoothed / lagged replays	Low-detail transforms populating the “plausible but lazy” region

Determinism and verifiability. The field seed is derived from the round’s pre-draw commitment (seed = SHA-256(commitment || field version)), so the house cannot regenerate a friendlier field and the player cannot anticipate field realizations. After settlement, anyone can re-derive all 1,500 paths from published code and reproduce the percentile and payout bit-for-bit. Generation plus scoring costs milliseconds. The generator is versioned; changes are announced and never retroactive.

Why relative ranking solves difficulty. The field experiences the same round — the same volatility, the same trendiness. If the round was easy, the field also scores high and the bar rises automatically. A given drawing quality maps to a similar percentile in calm and chaotic markets alike, which is precisely the fairness property an absolute score cannot provide.

5.3 Layer 3 — Payout

The multiplier is a published three-zone function of the percentile — a refund zone below the break-even percentile, an exponential profit zone above it, and a linear jackpot tail for the top of the field:

$$M(u) = m_{\min} + (1 - m_{\min})(u/u_{be})^{\alpha} \text{ for } u < u_{be}; \quad M(u) = \exp(g \cdot (u - u_{be})) \text{ for } u_{be} \leq u < u_J$$

with a linear ramp from $M(u_j)$ to $M_{\max}$ on $[u_j, 1]$. The curve is continuous, equals $1.0\times$ exactly at the break-even percentile, and the growth rate g is **solved numerically** so that the mean multiplier over the percentile range equals exactly $1 - h$ (Appendix B). Under the null hypothesis — a player no better than the field — the percentile is uniform, so the expected payout is exactly $1 - h$ per round, in every market regime. The house edge is a design constant, not an emergent property of tuning.

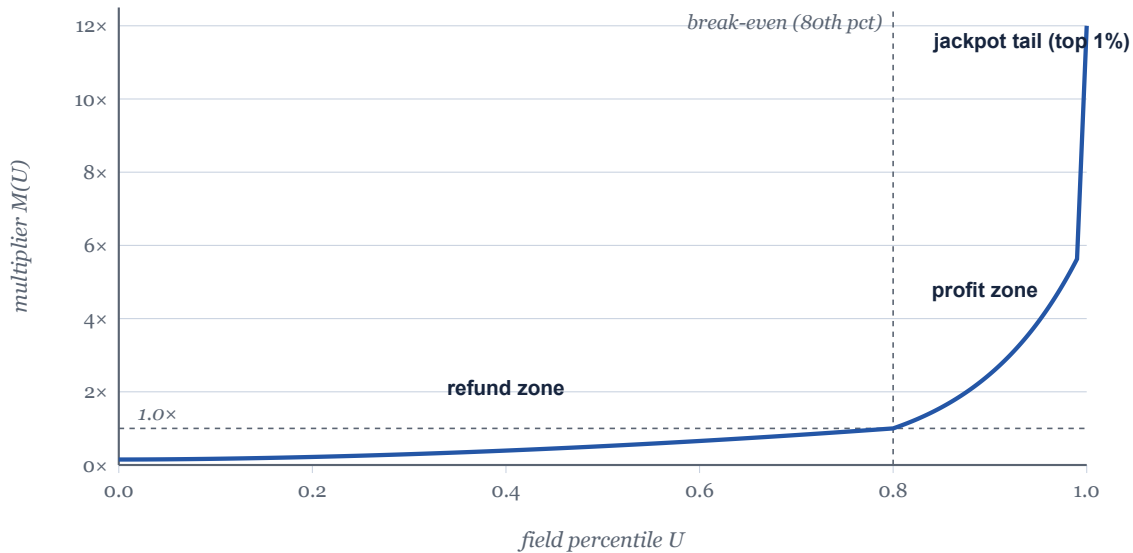


Figure 2 — The Standard payout profile ($h = 4\%$, break-even at the 80th percentile, cap $12\times$).

Three solved profiles span the frequency-versus-magnitude spectrum; the expected multiplier is invariant across them:

	Balanced	Standard (default)	Jackpot
House edge h	4%	4%	5%
Break-even percentile	0.75	0.80	0.88
Floor multiplier	$0.20\times$	$0.15\times$	$0.10\times$
Cap	$\approx 4.5\times$	$12\times$	$30\times$
Solved growth rate g	5.988	9.094	21.234
Mean multiplier (null)	0.9600	0.9600	0.9500
Std. dev. of multiplier	0.95	1.34	2.16
Rounds profitable (null)	25%	20%	12%

Standard profile, per \$100 stake:

Percentile	Multiplier	Payout	P&L
0.25	$0.26\times$	\$26	-\$74
0.50	$0.52\times$	\$52	-\$48
0.65	$0.74\times$	\$74	-\$26
0.80 (break-even)	$1.00\times$	\$100	\$0
0.90	$2.48\times$	\$248	+\$148
0.95	$3.91\times$	\$391	+\$291

Percentile	Multiplier	Payout	P&L
0.99	5.63×	\$563	+\$463
1.00 (best of field)	12.00×	\$1,200	+\$1,100

6. House Economics and Risk Management

6.1 Per-round accounting

For stake s , house profit per round is $s \cdot (1 - M(U))$ with mean $s \cdot h$ and a standard deviation known in closed form from the curve ($1.34 \cdot s$ for the Standard profile). Because both moments are exact under the null, bankroll mathematics requires no assumptions about player skill distributions.

6.2 Stake caps and bankroll sizing

With edge h and multiplier variance $\text{Var}[M]$, the full-Kelly stake fraction is approximately $h / \text{Var}[M]$ of bankroll. DTC operates at quarter-Kelly:

Profile	Full Kelly	Quarter-Kelly (operating cap)	Max stake at \$250k bankroll
Balanced	4.45%	1.11%	\$2,780
Standard	2.23%	0.56%	\$1,400
Jackpot	1.07%	0.27%	\$675

- Per-round liability $M_{\max} \cdot \text{stake}$ is reserved from the bankroll at entry; entries are rejected when total reserved exposure exceeds 25% of bankroll.
- Reserved exposure within any single horizon bucket is capped at 10% of bankroll, because same-horizon rounds settle on correlated market segments.
- Layered circuit breakers: 24h realized net loss beyond 1% of bankroll halves max stakes; beyond 2% pauses new entries; beyond 5% triggers emergency review. Because the multiplier variance is known, these thresholds correspond to computable z-scores — they fire on genuine anomalies, not normal variance.

6.3 Skill premium governance

The single residual economic risk is genuine, persistent player skill: a population whose mean percentile exceeds 0.5. This is exactly measurable — the realized margin $1 - \text{mean}(M)$ is tracked with confidence intervals overall, per horizon, and per cohort. If the margin compresses below $h/2$ for a sustained window, the curve is re-versioned with notice (higher break-even percentile or edge). If it reaches zero, stakes are floored and the field is upgraded: generators are added that capture whatever structure players are exploiting, which raises the bar for everyone honestly. Skilled players being paid is the product working; the governance loop keeps the cost known and bounded. All changes are versioned, announced, and never retroactive to open rounds.

7. Game Integrity

7.1 Commit-reveal

Before the player draws, the round payload is committed: anchor timestamp and price-source configuration, horizon, scoring version and parameters, field generator version and mixture, payout curve

version and solved g , and the seed-derivation rule. After settlement the payload is revealed, and settlement is a pure function of (commitment, drawn path, realized prices) that any third party can recompute.

7.2 Oracle

Settlement prices use an index construction: the median across at least three liquid venues, with 5-second TWAP windows centered on each one-minute sample, on a sampling schedule fixed in the commitment. A venue deviating more than 1.5% from the median is dropped for those samples; if fewer than two valid venues remain for more than 5% of samples, the round is voided and fully refunded. Path-scored games are more oracle-sensitive than endpoint markets — a single spoofed spike could create or erase structure — hence median, TWAP, and a conservative void policy.

7.3 Threat model

Attack	Defense
Play a naive archetype (flat line, trend, center-of-mass)	It is in the field — expected payout is exactly $1 - h$
Probe for scoring discontinuities with crafted paths	Layer 1 is smooth everywhere; money flows through rank, not raw score
Regime sniping (only play quiet or volatile rounds)	The field is conditioned on the same regime; percentile difficulty is invariant
Predict the field instead of the market	Field realizations depend on the committed seed, unknown pre-draw; mimicking its center is the naive-archetype case
House regenerates a friendlier field	Seed is bound to the pre-round commitment; settlement is reproducible by anyone
Multi-account spraying of random drawings	Each account independently faces expected payout $1 - h$; spraying has negative expected value
Draw with newer information than the anchor	Anchor locks at commitment; the draw window is 60–120 seconds and submissions are hash-bound inside it
Genuine forecasting skill	Paid on purpose within stake caps; monitored via skill-premium governance (§6.3)

8. Calibration and Validation Methodology

DTC treats calibration as a launch gate, not a polish step. All validation suites run against a historical dataset of 2.7 million BTC/USDT one-minute candles (January 2021 – March 2026) spanning a full bull run, a bear market, a recovery, and a halving cycle. The engine ships only when all of the following hold:

- **Zero-skill neutrality.** Every named zero-information strategy, played as a player over thousands of rounds per timeframe, earns a mean percentile within $[0.45, 0.55]$ and a mean multiplier within a tight band of $1 - h$. This directly asserts the economic property the system is built on.
- **Perceptual monotonicity.** Across graded distortion ladders (timing lag, amplitude damping, time warping, additive noise applied to the realized path), the mean percentile must decrease strictly with distortion, and a visually close drawing must lose head-to-head against a random drawing in fewer than 1% of rounds.
- **Skill responsiveness.** Partial-information oracles (direction-only, endpoint within a band, first half of the path) must earn percentiles materially above 0.5, increasing with information content — the

game must be winnable by insight.

- **Economic stress.** Monte Carlo across player mixes from 0% to 30% skilled: ruin probability below 0.1% over 100,000 rounds at launch bankroll under the stake and exposure rules of §6.
- **Human alignment.** The displayed accuracy score must correlate with players' self-assessed scores on collected feedback data, and improve on contested rounds across versions.
- **Engine invariants.** Bit-for-bit determinism, edge-case robustness, component correlation below 0.7, and third-party reproducibility of settlement from the commitment alone.

9. Roadmap

Phase	Scope	Status
1 — Scoring engine	Pure-function engine, field generator, payout solver, validation suites	Specified; v0.2 engine and 5-year backtest infrastructure operational
2 — Drawing experience	Freehand chart drawing, mobile touch, reveal animation	Built and validated in testing
3 — Game loop	Draw → rank → payout → replay; sandbox on historical data	Built; migrating to field-relative scoring
4 — Live infrastructure	Wallet auth (Base), on-chain commit-reveal, oracle integration, bankroll vault, risk controls	In design
5 — Depth	Leaderboards (percentile-fair across regimes), tournaments, PvP fields (rank against real players), additional assets	After live economics are proven

10. Compliance and Responsible Use

DTC is a speculative entertainment product wrapped in a market-native interface, and is presented as such. The live product requires jurisdictional gating and legal review before real-money rollout, age gating, self-exclusion, and per-user session, stake, and loss limits. The field-relative design enables unusually honest disclosure: the product can truthfully state that the expected return of play without predictive skill is exactly $1 - h$ per round, because that number is enforced by construction rather than estimated.

11. Token Position

DTC does not require a token. The loop — draw, rank against the field, settle — stands alone on stablecoin rails. Any future token design would be subordinate to the product, bounded in utility, and deferred until product-market fit.

12. Risks

Risk	Mitigation	Residual
Field mis-specification (a naive strategy escapes the neutrality band)	Five-year backtest neutrality suite; versioned field upgrades	Low

Risk	Mitigation	Residual
Persistent player skill	Exact measurement, governance ladder, stake caps	Medium — also the product working
Correlated wins in strong trends	Trend generators inside the field raise the bar in trend rounds; horizon-bucket exposure caps	Low–medium
Oracle manipulation	Multi-venue median, TWAP, void-and-refund policy	Low
Percentile feels opaque (“who is the field?”)	Field swarm visualization, inspectable rival paths, published generator code	UX risk; testable
Regulatory classification	Staged launch, legal review, geo-gating	Jurisdiction-dependent

13. Conclusion

Crypto users already think in chart paths, already make scenario calls, and already want the feeling of reading the market without the complexity of leveraged execution or the poverty of binary input. Draw The Chart turns that existing behavior into a game with a precise economic core.

The field-relative architecture changes the question the game asks. Not “*how close was your drawing on an absolute scale that shifts with every regime?*” but “*did you beat the field of every naive strategy, on this round, under the same conditions?*” That question is fair by construction (the field absorbs difficulty), legible by construction (“top 14% of the field”), exploit-resistant by construction (every naive strategy is in the field), and solvent by construction (the payout curve integrates to exactly $1 - h$). What remains is execution against the validation gates of Section 8.

Appendix A — Default Parameters

Layer	Parameter	Default
Similarity	Resample points N	120
Similarity	DTW band w	12 samples (10% of horizon)
Similarity	Shape decay λ_s	2.2
Similarity	Direction scale levels / decay	3 / 2.5
Similarity	Level decay λ_l	1.2
Similarity	Component weights (shape/direction/level)	50 / 30 / 20
Similarity	Volatility floor coefficient	$0.5 \cdot \sigma$ typical (timeframe)
Field	Field size B	1,500
Field	Mixture	40 / 20 / 15 / 10 / 10 / 5 (\$5.2)
Field	Bootstrap block length	horizon / 8
Field	EWMA volatility λ	0.94
Payout	Profile (Standard)	$h = 0.04, u_{be} = 0.80, m_{min} = 0.15, \alpha = 1.8, u_j = 0.99, M_{max} = 12, g = 9.0944$

Layer	Parameter	Default
Risk	Stake cap	0.56% of bankroll (quarter-Kelly)
Risk	Total / per-horizon reserved exposure	25% / 10% of bankroll

Appendix B — Payout Curve Mathematics

The mean multiplier over the percentile range decomposes into closed-form pieces:

$$\text{refund: } u_{be} \cdot (m_{min} + (1 - m_{min}) / (\alpha + 1))$$

$$\text{profit: } (\exp(g \cdot (u_J - u_{be})) - 1) / g$$

$$\text{jackpot: } (1 - u_J) \cdot (\exp(g \cdot (u_J - u_{be})) + M_{max}) / 2$$

The growth rate g is solved by bisection so the sum equals $1 - h$; the sum is monotone increasing in g , so the root is unique. For the Standard profile, $g = 9.0944$ gives a mean multiplier of 0.9600 with standard deviation 1.339 (verified by 400,000-point numeric integration). Because the live percentile is discrete (resolution $1/B$), g may equivalently be solved against the grid sum directly, making the expected payout exact to the cent. The house profit per unit stake has mean h and variance $\text{Var}[M]$, from which the quarter-Kelly stake cap $h / (4 \cdot \text{Var}[M])$ follows.

Appendix C — Timeframes and Drawing Constraints

Timeframe	Horizon	Display interval	Control points
15m	15 minutes	1m	8
1h	60 minutes	1m	12
6h	360 minutes	5m	16
24h	1,440 minutes	15m	16
7d	10,080 minutes	1h	16

First point fixed at the anchor; last point at the horizon end; strictly increasing time coordinates; minimum spacing of horizon/ $K/2$ between control points; maximum slope bounded at $3 \times$ the historical maximum hourly move.